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clockplot: An R package to Plot Event Times on a 24-Hour Clock

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Software

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Summary

Visualization is a fundamental component of statistical analysis and prediction, enabling clear data description and informed model selection. While various methods exist for visualizing event data, there is a notable gap in representing events on a clock face. In this paper, we introduce `clockplot`, an R package for plotting timestamped events on a 24-hour circular clock face. This approach offers a powerful way to perceive the precise timing of events and facilitates intuitive comparisons with other events. The `clockplot` is particularly useful for analyzing daily patterns, event clustering, and temporal gaps. For event times, it is significantly more revealing than conventional visualizations like bar or pie charts. This package also generalizes the clockplot approach to create cyclic charts for other time frames, including weekly and monthly cycles. This functionality enables effective event planning and pattern analysis across multiple periods, providing a new and insightful tool for temporal data analysis.

Statement of Need

Visualizing temporal patterns in timestamped event data presents unique challenges, particularly when events exhibit cyclical behavior over 24-hour periods. Traditional linear timelines (e.g., using `ggplot2::geom_point()` with time on the x-axis) have several limitations for this type of analysis:

1. **Circular continuity is broken**: The natural continuity between 23:59 and 00:00 is not preserved, making it difficult to see patterns that span midnight.
2. **Density patterns are hard to discern**: Clustering of events around specific times of day is less visually apparent on linear axes.
3. **Multiple days of data become cluttered**: Plotting several days of data on a linear timeline creates overlapping points that obscure daily patterns.

Available Software

While several R and Python packages offer circular or radial visualization capabilities, they are either too general-purpose or not specifically designed for timestamp data:

- **General circular plotting**: Packages like `circular` and `plotrix` provide basic circular plotting (Agostinelli & Lund, 2025; Lemon, 2006), but require extensive data transformation for timestamp visualization.
- **Polar coordinates**: `ggplot2` with `coord_polar()` can create circular plots, but requires manual conversion of timestamps to radians and lacks specialized aesthetics for event data (Wickham, 2016).
- **Specialized but limited**: Packages like `chron` and `lubridate` handle time data but don't provide circular visualization methods (Grolemund & Wickham, 2011).

- **CircadiPy**: deals with rhythmic data, but does not create circular visualizations (Carvalho-Moreira et al., 2024).

The package provides specialized visualization methods for multivariate data including **colored point aesthetics** (mapping variables to hue/size/shape) and **modifiable clock hand lengths** (representing magnitude or intensity at each timestamp). These features enable clear differentiation of multiple variables on the same 24-hour clock face while maintaining intuitive interpretation of temporal patterns.

Acceptable Data Type

`clockplot` handles two primary data types:

1. Timestamped Event Data

For analyzing patterns in event occurrences, such as website visits, system logs, or biological rhythms:

Table 1: Sample Univariate Timestamps

ID	Time	ID	Time
1	21:12:27	5	03:16:57
2	02:20:09	6	03:35:33
3	16:46:37	7	21:44:36
4	08:57:52	8	12:45:15

2. Multivariate Time Series Data

For visualizing continuous measurements at specific times, such as environmental monitoring or physiological data:

Table 2: Sample Multivariate Timestamps

Time	Temperature	Time	Temperature
21:12:27	20.1°C	03:16:57	21.4°C
02:20:09	20.2°C	03:35:33	20.6°C

Features

The following code creates a circular visualization of USA accidents:

```
library(tidyverse)
library(clockplot)
acdt <- read.csv("https://raw.githubusercontent.com/mahmudstat/open-analysis/main/data/u
acdt %>% ggplot(aes(Time, Humidity...))+
  geom_bar(stat = "identity")+
  labs(x = "Time",
       y = "Humidity",
       title = "USA Accidents Time and Corresponding Humidity")
```

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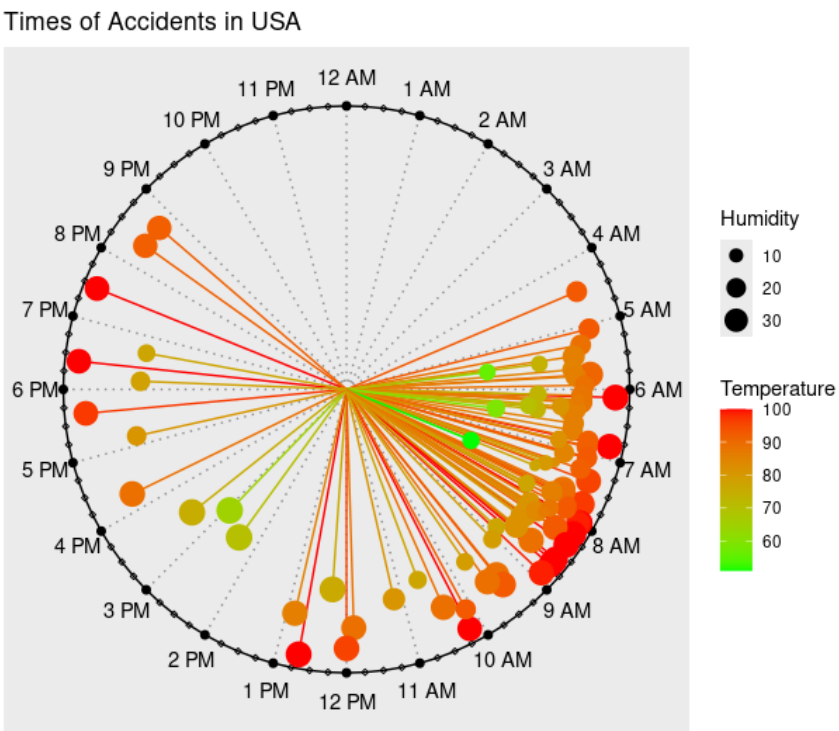

Figure 1: USA Accidents Time and Corresponding Humidity

Figure 1 reveals several important patterns:

1. **Clear temporal clustering**: Accidents peak during morning rush hours (5-9 AM)
2. **Critical gaps**: Noticeably fewer accidents occur between 9 PM to 4 AM
3. **Multivariate encoding**: Each point encodes two additional variables:
   - **Point size** represents humidity levels
   - **Point color** represents temperature (red = warmer, green = cooler)
4. **Cross-midnight continuity**: The natural connection between late-night (11 PM) and early-morning (1-2 AM) accidents is preserved

The plot enables immediate identification of high-risk periods while simultaneously showing how environmental factors correlate with accident frequency—a multidimensional insight difficult to achieve with traditional linear timelines.

Similar conclusions can be drawn from Figure 2.

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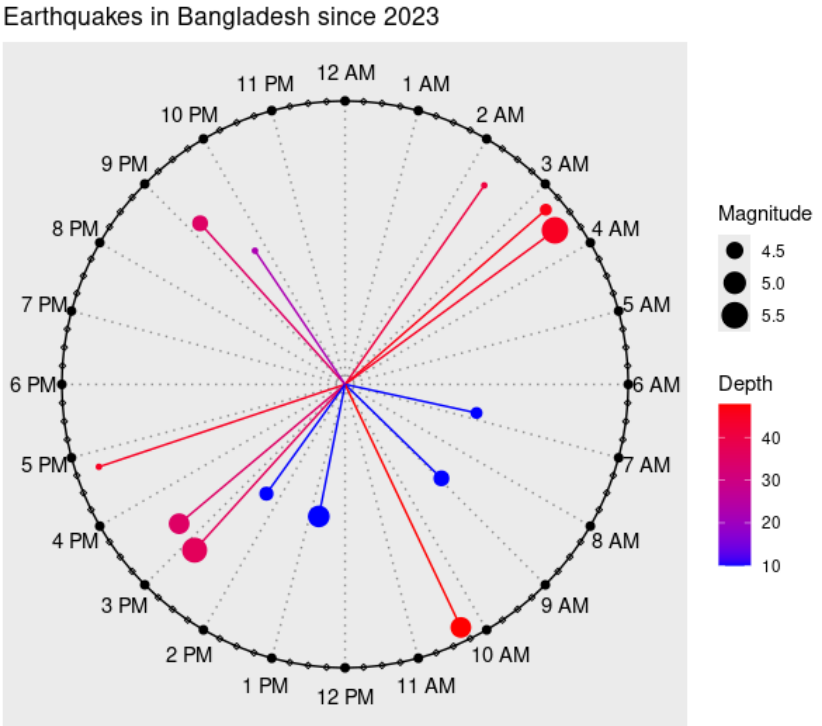

Figure 2: Recent Earthquakes in Bngladesh

The visualization reveals significant temporal patterns in seismic activity:

1. **Temporal gaps**: The most prominent gaps in earthquake occurrence are observed between **5 PM to 9 PM** and **10 PM to 2 AM**, with no recorded earthquakes during these periods in the sample data.
2. **Multivariate encoding**: The plot encodes two seismic parameters:
   - **Clock hand length** represents earthquake depth (longer hands = deeper epicenters)
   - **Point size** corresponds to magnitude (larger points = stronger earthquakes)
3. **Analytical insights**: This representation enables simultaneous analysis of **when** earthquakes occur and their **physical characteristics** (depth and magnitude), revealing potential correlations between time of day and seismic parameters.

This exemplifies the package's capability for multivariate temporal visualization, where traditional linear timelines would require separate plots for each variable.

Figure 3 plots qualitative data, showing not only when SMS messages were received but also who sent them.

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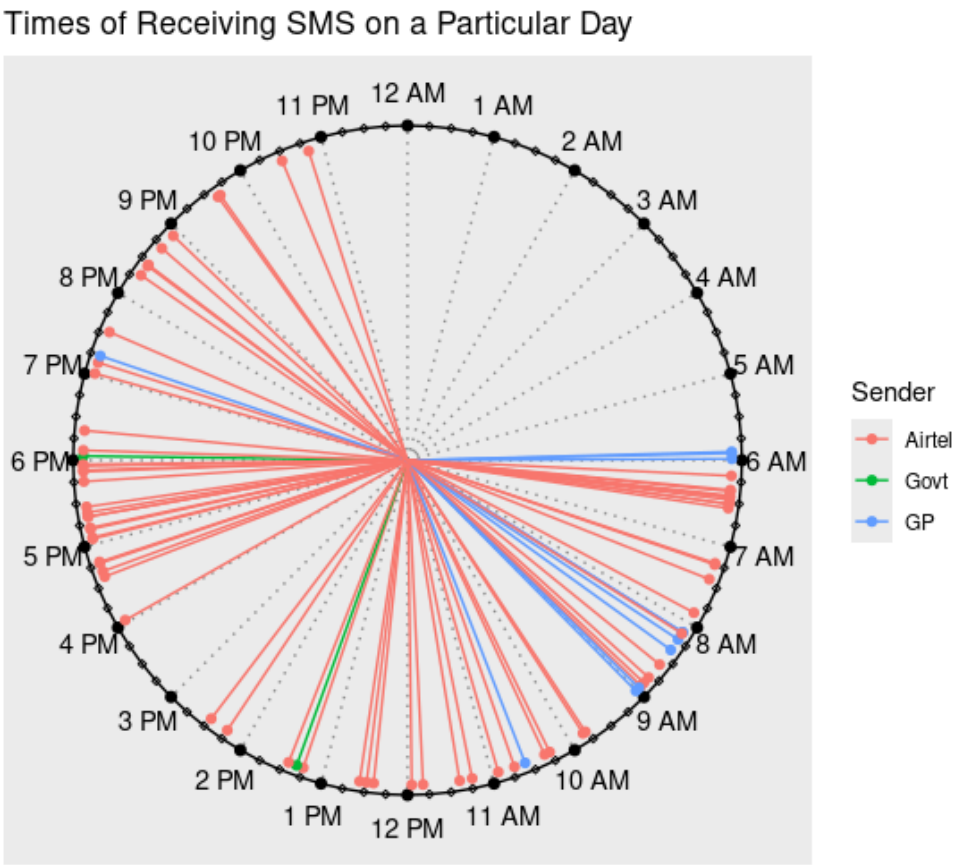

Figure 3: Timestamps of Incoming SMS and Senders

Usage

Complete documentation with all arguments, options, and examples is available at mahmudstat.github.io/clockplot.

Acknowledgments

The example analyses use data from public repositories including the US Accidents Dataset and earthquake data from the United States Geological Survey (USGS).

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Conflict of Interest

The authors declare that they have no conflicts of interest related to this work.

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